Jaume Puig

Space Radiobiologist and Engineer

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PROFESSIONAL EXPERIENCE

Oct 2024 to Present

CTO & Co-Founder

Barding Industries

- Architecting the technology platform: engineered biology that recovers critical minerals, including rare earth elements, from ores and waste streams.

Jun 2024 to Present

Communications Coordinator & Member, Microbes & Hardware AWGs

NASA Open Science Data Repository

- Contributing to open-science space microbiology through omics data reanalysis, pipeline optimization, and methods standardization.
- Co-authoring the SPLASH Database paper alongside AWG principal investigators.
- Managing internal communications and inter-subgroup coordination across the AWG.

Feb 2021 to Present

Judge, Mentor & Community Member

iGEM Foundation

- Executive Judging Committee Member (2026 to Present): guiding the evolution and quality of judging across the competition, from judging philosophy, standards, and rubrics to training incoming judges and ensuring fairness and consistency.
- Space Network Member (2024 to Present): advising teams on space exploration and astrobiology applications.
- Judge (2024 to Present): evaluating international synthetic biology projects.
- Mentor (2022 to Present): technical guidance across experimental design and modeling.
- Ambassador for Europe (2022 to 2024): promoted synthetic biology across Europe.

May 2024 to Present

STEAM Educator

AulaSPUTNIK

- Designing and delivering science workshops for children, and managing the organization's website.

Jun 2024 to Jan 2025

Research Fellow, Crew 2: The Illuminator's Class

BioAstra

- Analyzed pharmacogenomic and nutritional genomics datasets from spaceflight missions to advance commercial astronaut health research.
- Contributed to protocol development for the Space Omics and Medical Atlas (SOMA) framework.

Apr 2023 to Jan 2024

Space Biotechnology Technician

Yuri GmbH

- Conducted microbial culture experiments and multi-omics analysis for ISS payload projects using automated modular bioreactors.
- Managed laboratory facilities, hardware testing, and data interpretation across multiple concurrent payload programs.

ACADEMIC BACKGROUND

2025 to Present

Ph.D. in Biomolecular Sciences
Laurentian University / SNOLAB

- Major: Radiation Biology.
- McLaughlin Graduate Scholarship recipient.
- Teaching Assistant.

2022 to 2023

M.Sc. in Multidisciplinary Research in Exp. Sciences
BIST / UPF

- Major: Low-Background Radiation Omics. Minor: Ancestral Gene Resurrection.
- Lead organizer of ScienceHack 2023 (BIST-MMRES), a science hackathon with 360+ participants.

2016 to 2022

B.Eng. in Biomedical Engineering
Universitat Pompeu Fabra (UPF)

- Major: LEO Radiation Biotolerance.
- iGEM 2020: Gold Medal, Best New Basic Part, Best Wiki. Nominated for Best Therapeutics Project, Best Poster, and Best Presentation.

PUBLICATIONS & PRESENTATIONS

- **Publication.** npj Microgravity: spaceflight pharmacogenomics and protocols for SOMA (2025).
- **Publication.** Frontiers in Systems Biology: intein-mediated thyroid hormone biosensors (2024).
- **Publication.** Frontiers in Microbiology: DNA protection strategies under simulated LEO radiation (2021).
- **Presentation.** Radiation Research Society (RRS) Annual Meeting, October 2026.
- **Presentation.** International Astronautical Congress (IAC) 2023, Baku: skin microbiome radiation biotolerance.
- **Presentation.** AGU / NASA AbSciCon 2022, Atlanta: DNA protection in low Earth orbit.
- **Presentation.** Europlanet Science Congress 2022, Granada: DNA protection in low Earth orbit.